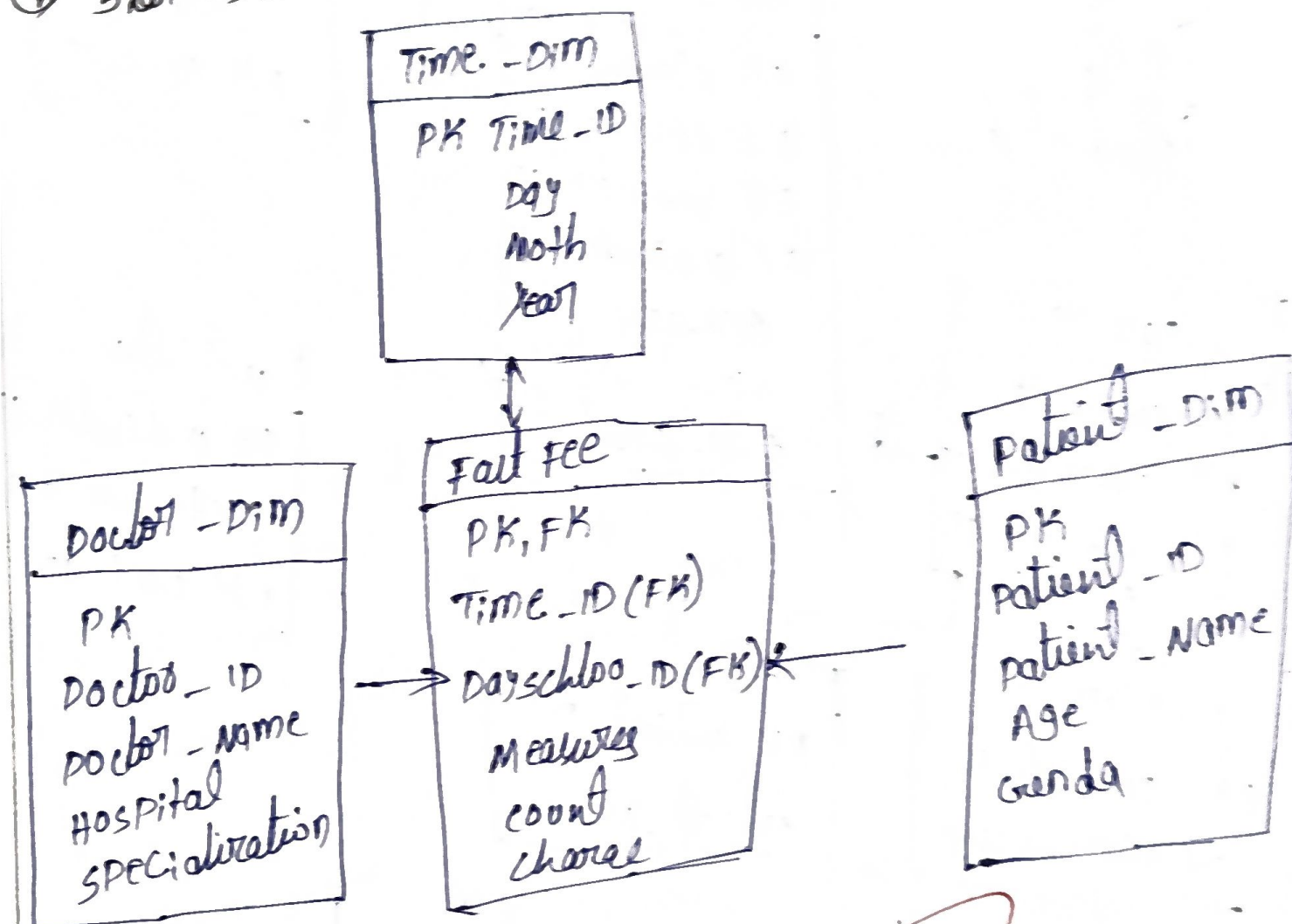


data warehouse: Doctor - patient visits

(a) Three popular schemas used in data warehouse are

- 1) star schema
- 2) snowflake schema
- 3) fact constellation schema

(b) star schema diagram



(c) SQL query

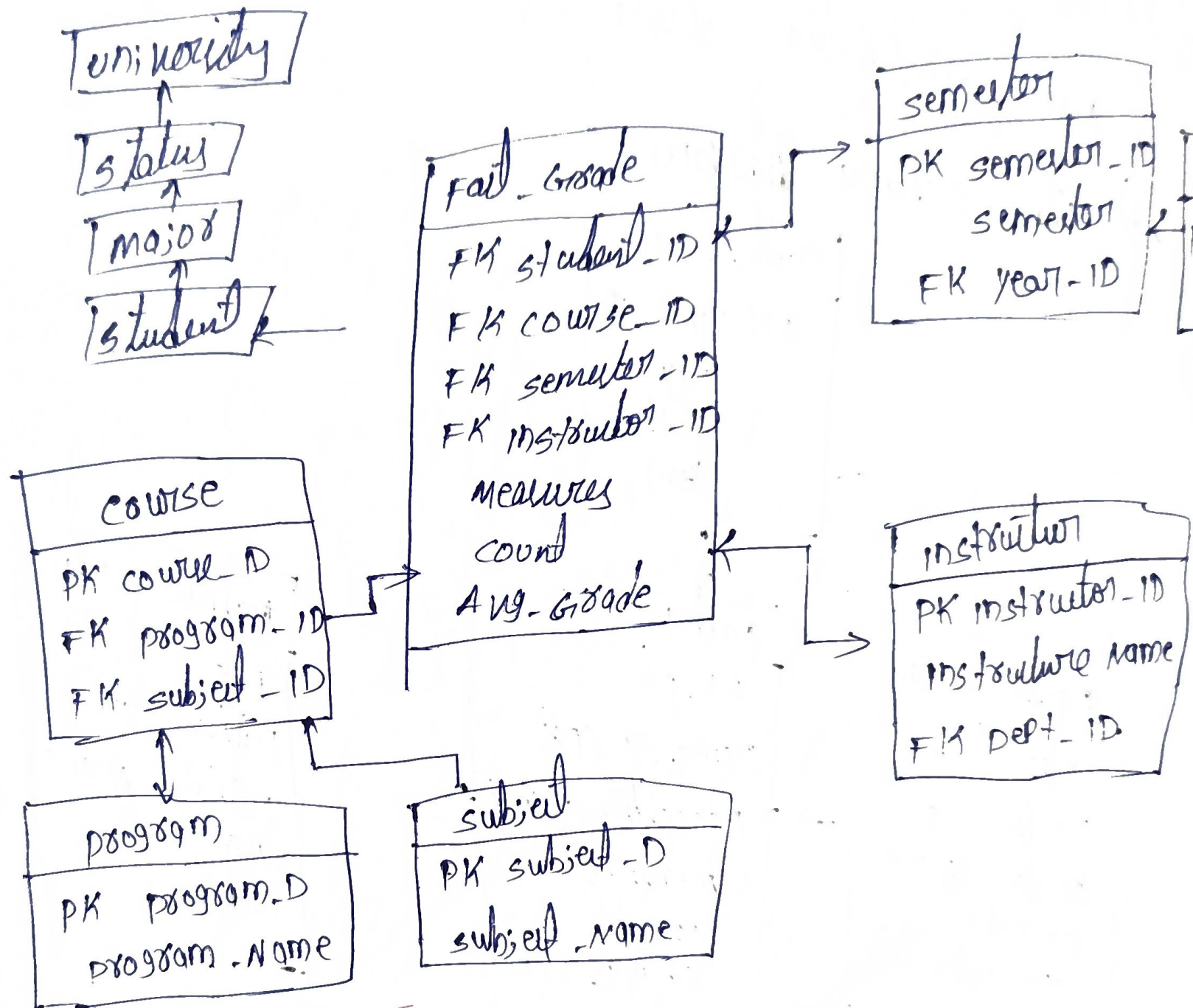
```
SELECT
  doctor
  patient
  sum (count) AS Total - visits
  sum (charge) AS Total - charge
```

FROM fee

GROUP BY doctor, patient;

2. Big university data warehouse

(a) snowflake schema diagram



(b) OLAP operations:

~~starting~~ cuboid: [students, course, semester, instructor]
operations:

1. slice
* select only CS course
course = 'CS'

(i) no. of cuboids

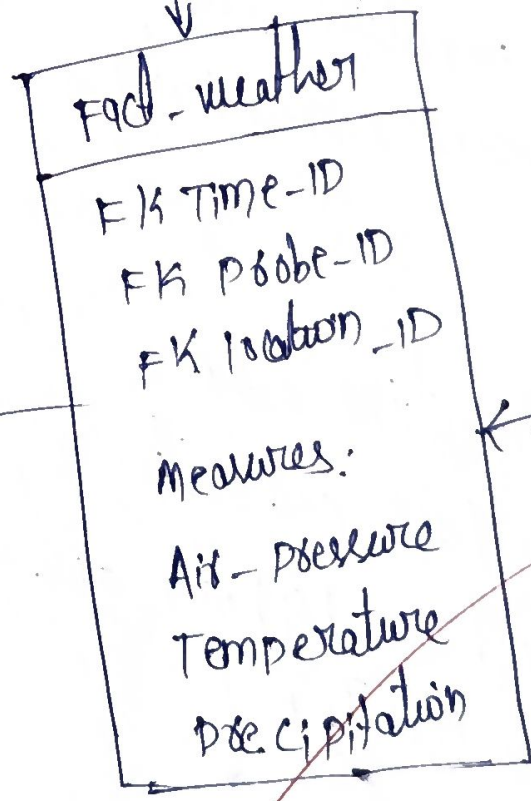
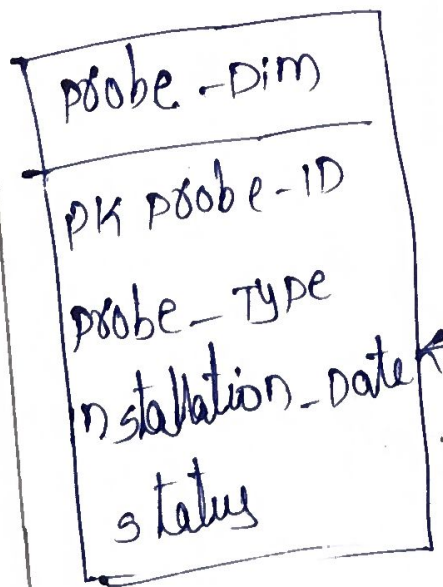
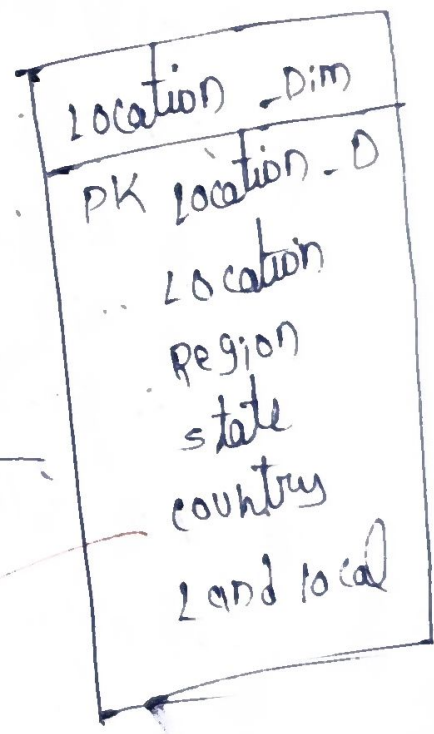
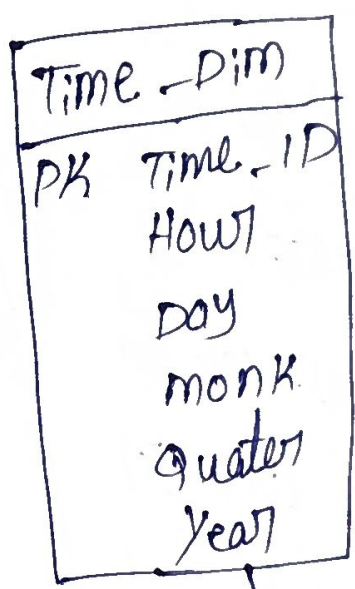
formula = cuboids = $\prod (L_i)$

where, L_i = no. of levels in each dimension

each dimension has 5 levels

$5 \times 5 \times 5 \times 5 = 625$

Regional weather Bureau data warehouse star schema
design



Advantages:

- analyze temperature by region / year
- compare land vs ocean weather
- perform O/NP operations efficiently

4. Data cube problem

Given n dimensions
each dimension has distinct value
No hierarchy

(a) max. cells in Base cuboid
every combination exists

(b) minimum cells in Base cuboid
only one non-empty cell. $\square$

(c) max cells in entire cube
each dimension contributes
($P+1$)

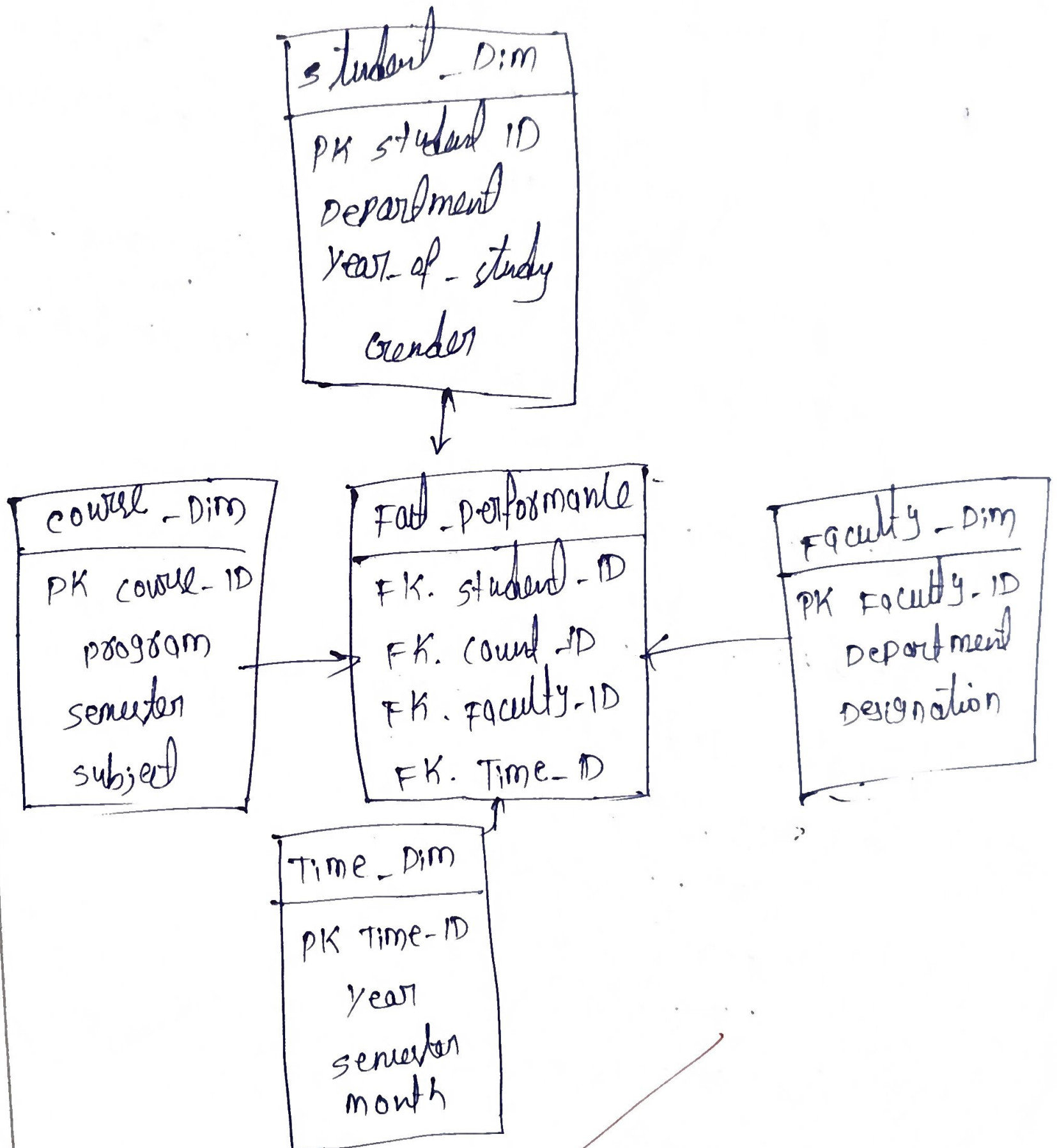
$$\therefore (P+1)^n$$

(d) min cells in entire cube

→ if only one base cell exists
every cuboid contributes one aggregate cell

NP. of cuboids $\square 2^n$

university Academic performance data warehouse
c9) star-schema design



(b) OLAP operations

step	operation	description	Result
1	Roll-up	student $\rightarrow$ department	[department, course, Faculty, Time]
2	Roll-up	Time: month $\rightarrow$ semester $\rightarrow$ year	
3	Roll-up	course $\rightarrow$ All	[department, All, Faculty, year]
4	Roll-up	Faculty $\rightarrow$ All	[department, All, All, year]

Result: [department, All, All, year]

(c) Total no. of cuboid

Formula: cuboids = $L_1 \times L_2 \times L_3 \times L_4$

Given: student = 4 levels

course = 3 levels

Faculty = 3 levels

Time = 5 levels

$$4 \times 3 \times 3 \times 5 = 180$$

$\therefore$ 180 cuboids

mg